

System Plan: UniConnect

1. MVP Screens & User Flow

The MVP consists of 3 main screens:

1. **Login/Register:** The entry point for authentication.
2. **Dashboard (The Hub):** The main screen divided into several tabs:
 - **Resources:** Course related files and more uploaded by teachers.
 - **Announcements:** Global messages from teachers or class representatives.
 - **Chat:** Real-time communication for all group members.
 - **Schedule:** Timetable and assignments view for students; assignment creation for teachers.
 - **Event Creation:** Tool for students to organize study sessions or activities.
 - **Management Screen (Teachers/Admins only):** Roster management and promotion of class representatives.
3. **Profile & Settings:** User info, updates, app settings, and logout.

Primary User Flow:

- The user logs in with credentials and receives a token.
- They are redirected to the Dashboard to see their specific class menus (e.g., "Group A").
- From there, they can access the Chat, Schedule, or Resources tabs.

2. Module/Component Breakdown

The application uses a separated Client-Server architecture.

Frontend (Client - HTML/CSS/JS):

- **AuthManager:** Manages JWT storage (LocalStorage) and redirections.
- **ChatWidget:** Manages WebSocket connections (ws://) and message display.
- **TabNavigator:** Handles switching between Dashboard UI tabs.
- **APIHandler:** A wrapper for REST API fetch calls (GET, POST, PATCH, DELETE).

Backend (Server - FastAPI Python 3.12):

- **Auth Router:** Manages tokens (OAuth2) and registration.
- **SocketManager (Service):** A singleton managing active WebSocket connections.

- **CRUD Routers:** Modules for Users, Groups, Events, Assignments, Schedules and Ressources.
- **Database Core:** Handles SQLAlchemy sessions and models.

3. Minimum Data Model (SQLite)

Data is stored server-side using SQLAlchemy:

- **User:** { id, email, password_hash, role, first_name, last_name, group_id }
- **Group:** { id, name, schedule_path }
- **UserGroup:** (Association Table) { user_id, group_id }
- **Message:** { id, content, sent_at, user_id, group_id }
- **Event:** { id, title, description, date, location, type, group_id }
- **Assignment:** { id, title, description, due_date, group_id }
- **Resource:** { id, title, file_path, file_type, size, uploaded_at, user_id, group_id, category }
- **Candidate:** { id, first_name, last_name, manifesto, group_id, photo_url }
- **Vote:** { id, user_id, candidate_id, group_id }

4. Data Flow for MVP Features

- **User Authentication:** Frontend sends credentials via POST; Backend verifies and returns a JWT with the user's role. Frontend saves the token in localStorage.
- **Class Group Management:** Professors send group or roster updates via POST/PATCH; Backend validates the professor role before updating the Group and User tables.
- **Group Messaging:** Text content is sent via WebSocket; Backend persists the message to the Message table and broadcasts it to the group via the SocketManager.
- **Official Announcements:** Professors or Delegates submit posts via POST; Backend verifies permissions.
- **Timetable Access:** Frontend requests the weekly schedule via GET; Backend retrieves official course data from the Group record for display on the grid.
- **Study Event Organization:** Students submit session details via POST; Backend saves the data to the Event table, which the Frontend then overlays on the timetable grid.
- **Election & Voting:** Student sends a candidate ID via POST; Backend checks the Vote table for the unique constraint (one vote per user/group) before saving.

5. Technology Approach & Constraints

- **Tech Stack:** Python 3.12, FastAPI, SQLAlchemy, and SQLite for the backend. The frontend uses Vanilla JavaScript with plain HTML5 and CSS3.
- **Protocols:** HTTPS/1.1 for CRUD operations and WSS for real-time chat.
- **Constraints:** No "Forgot Password" feature (manual reset by Admin). Files are stored locally on the server disk rather than in the cloud.

UniConnect: Technical Architecture Diagram (Client-Server with WebSockets)

Team UniConnect Devs - LA1332 System Plan

